Arithmetic Quantum Cat Maps:
Exact Fourth-Moment Reductions, Square Cyclotomic Resultants, and the Remaining BGS Frontier

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Abstract

We develop an exact finite-field reduction of a fourth-moment problem arising in arithmetic quantum cat maps and isolate the arithmetic obstruction that remains before any microscopic Bohigas–Giannoni–Schmit (BGS) statement can be contemplated. The pair-key geometry gives an exact constrained-pair count $(K-1)(K-2)$ and a corrected fourth-moment difference distribution

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0}.$$

A Cayley transform turns the mixed term into the nonzero additive-difference energy of a Mobius image of a multiplicative subgroup. For odd prime subgroup order K, Chowla's rational independence theorem for cotangents gives a uniform characteristic-zero classification of all equal-difference collisions. We then prove a new structural identity for every cleared collision polynomial P:

$$\mathrm{Res}(\Phi_K, P) = K^3M^2, \qquad M \in \mathbf{Z}.$$

This improves the elementary no-collision cutoff by a square root. An explicit collision slice satisfies

$$\mathrm{Res}(\Phi_K, X^2+3X+1) = F_K^2,$$

so the square factor can grow exponentially and a polynomial bound for the whole resultant norm is impossible. Nevertheless this exceptional slice is arithmetically sparse: for $S_N = \{\pm L_n/F_n : 1 \le n \le N\}$ its exact nonzero difference energy is

$$E^*(S_N) = 2N(4N-3)+32 \qquad (N \ge 3).$$

A complete exact certificate at the 28-digit prime $p = F_{131} = 1066340417491710595814572169$ reproduces the same excess 32 in the finite Cayley set. Thus the correct growing-K mixed-term target is a uniform *energy* bound, not elimination of every exceptional collision. The off/off elliptic-family estimate remains a theorem draft, and no result in this paper proves microscopic BGS/GOE eigenphase statistics.

1 Scope and evidence discipline

Throughout, p is an odd prime and $H \le \mathbf{F}_p^\times$ is a multiplicative subgroup of order K. Division is in $\mathbf{F}_p$. We distinguish:

- **Exact**: a current proof with explicit hypotheses;
- **Finite certificate**: an exhaustive or kernel-checked statement at explicit finite parameters;

- **Finite regression**: an implementation check that supports, but does not prove, a universal statement;
- **Numerical**: descriptive floating-point evidence only;
- **Draft/Open**: a named closure condition remains.

The distinction between moment information and microscopic spectral statistics is essential. Arithmetic cat-map literature contains strong matrix-element, eigenfunction and quantum-ergodicity results [1, 2, 3, 4, 5], but these are not interchangeable with a local eigenphase law. Conversely, rigorous BGS-type results are known for special model classes such as mixing quantum graphs under their own large-graph hypotheses [6, 7]; they do not transfer to the arithmetic family treated here without a new argument.

2 Exact pair-key geometry

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Off-diagonal pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every such ordered pair (u, v) recovers a unique admissible off-diagonal (x, y). The diagonal $x = y$ collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. From $x = uy$ and $x - 1 = v(y - 1)$, $(u - v)y = 1 - v$, which gives the inverse formulas and then the formula for d. The diagonal statement is immediate. □

Corollary 2.2 (Exact constrained-pair count)**.** *If the two constraints are $u, v \in H$, the number of admissible off-diagonal ordered pairs is*

$$(K - 1)(K - 2).$$

Proof. Choose an ordered pair of distinct elements of $H \setminus \{1\}$ and apply Proposition 2.1. □

Status: Exact.. Independent Python/C++ checks and a Lean certificate at $(p, K) = (53, 13)$ guard the implementation, but are not the source of the general proof.

3 Fourth-moment coupling and finite Fourier identities

Give each collapsed key $r = (u_r, v_r, d_r)$ multiplicity n_r, equal to one off the diagonal and $p - 2$ at $(1, 1, 0)$. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \mathbf{1}_H(v_r v_s) \mathbf{1}_{d_r + d_s = t}. \tag{3}$$

Let $O(t)$ be the off/off contribution and

$$C_H(t) = \#\left\{(u, v) \in (H \setminus \{1\})^2 : u \neq v, \ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

Then the diagonal bookkeeping is exactly

$$B(t) = O(t) + 2(p - 2)C_H(t) + (p - 2)^2 \mathbf{1}_{t=0}. \tag{4}$$

The coefficient 2 is load-bearing: diagonal/off and off/diagonal are distinct ordered classes.

Proposition 3.1 (Exact total coupled count)**.**

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K.$$

Let $S = H^\perp$ be the multiplicative-character annihilator, so $m = |S| = (p-1)/K$. In the maintained twisted-Jacobi convention,

$$\sum_{\chi,\psi\in S} |J(\chi,\psi;q)|^4 = m^2\Sigma_4(q), \qquad \Sigma_4(q) = \sum_{t\in\mathbf{F}_p} B(t)e_p(qt). \tag{5}$$

Writing $V = \sum_t B(t)$ and $Z = B(0)$,

$$\mathbb{E}_{q\neq 0}\Sigma_4(q) = \frac{pZ - V}{p-1}, \qquad \mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = p\,\mathrm{Var}_{t\neq 0}\,B(t). \tag{6}$$

These follow directly from finite Fourier orthogonality and Parseval; no independence heuristic enters.

4 Cayley transform and additive energy

Put

$$c(h) = \frac{h+1}{h-1}, \qquad h \in H\setminus\{1\}.$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u)-c(v)}. \tag{7}$$

Thus for $A = c(H\setminus\{1\})$ the nonzero multiplicities of C_H are the nonzero difference multiplicities of A, up to the permutation $x \mapsto 2/x$. Consequently

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{8}$$

Since $c(h^{-1}) = -c(h)$, $A = -A$ and $0 \notin A$. With $n = |A| = K-1$, identity and inversion-swap alone force the nonzero-energy baseline

$$n(2n-3) = (K-1)(2K-5). \tag{9}$$

A separate exact matching audit of all $6! = 720$ integer systems obtained by clearing and matching Cayley collisions gives ranks

$$30 \text{ (rank 2)}, \qquad 240 \text{ (rank 3)}, \qquad 450 \text{ (rank 4)},$$

and an extra rank drop can occur only in characteristic 2. Complete fixed-K cyclotomic enumeration currently gives

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\}. \tag{10}$$

These are exact computer-assisted finite statements, not a growing-K theorem.

5 Prime-order characteristic-zero rigidity

Let K now be an odd prime and $\zeta = e^{2\pi i/K}$. For $1 \le j \le K-1$,

$$c(\zeta^j) = -i \cot \frac{\pi j}{K}. \tag{11}$$

Chowla proved that

$$\cot \frac{\pi a}{K}, \qquad 1 \le a \le \frac{K-1}{2},$$

are linearly independent over $\mathbf{Q}$ [8]; Wang states this prime case explicitly and extends it [9].

Lemma 5.1 (Cross-polytope equal differences)**.** *Let x, y, u, v be vertices of $\{\pm e_1, \dots, \pm e_m\}$ with $x \neq y$ and $u \neq v$. If $x - y = u - v$, then either $(u, v) = (x, y)$ or $(u, v) = (-y, -x)$.*

Proof. The equality is $x + v = u + y$. If the common sum vanishes, then $v = -x$ and $u = -y$. Otherwise a nonzero sum of two signed basis vectors uniquely determines its unordered pair of summands. The alternative pairing would force $x = y$, excluded. □

Theorem 5.2 (Prime-order collision rigidity)**.** *Every characteristic-zero equal-difference collision in $\{c(\zeta^j) : 1 \le j < K\}$ is either identity or central inversion-swap.*

Proof. Use (11), replace each cotangent with the corresponding signed basis vector via Chowla's rational independence, and apply Lemma 5.1. □

6 Square cyclotomic resultants

For nonzero residues $i, j, r, s \pmod K$, with $i \neq j$ and $r \neq s$, choose integer lifts and define the cleared collision polynomial

$$\begin{aligned} P(X) = {} & X^{r+s+i} - X^{r+s+j} - X^{r+i+j} + 2X^{r+j} - X^r \\ & + X^{s+i+j} - 2X^{s+i} + X^s + X^i - X^j. \end{aligned} \tag{12}$$

Direct denominator clearing gives

$$2P(X) = \big(c(X^r) - c(X^s) - c(X^i) + c(X^j)\big) \prod_{a \in \{r,s,i,j\}} (X^a - 1). \tag{13}$$

The coefficient ℓ^1 norm satisfies $\|P\|_1 \le 12$.

Lemma 6.1 (Cubic zero and reciprocity)**.** *Writing $T = i + j + r + s$, one has*

$$P(X) = (X-1)^3 Q(X), \qquad Q \in \mathbf{Z}[X],$$

and

$$Q(X^{-1}) = X^{3-T} Q(X).$$

Proof. In (13), cancel the denominator of each Cayley term separately; every summand retains three factors $X^a - 1$, proving the cubic zero. Furthermore $c(X^{-a}) = -c(X^a)$ and $X^{-a} - 1 = -(X^a - 1)/X^a$. The four denominator factors contribute positive sign and total power X^{-T}, while the Cayley-difference contributes one minus sign. Hence $P(X^{-1}) = -X^{-T} P(X)$, which gives the formula for Q. □

Theorem 6.2 (Square-resultant theorem)**.** *For every odd prime K and every admissible collision quadruple,*

$$\operatorname{Res}(\Phi_K, P) = K^3 M^2 \tag{14}$$

for some $M \in \mathbf{Z}$. For a non-forced characteristic-zero collision, $M \neq 0$.

Proof. Let $L = \mathbf{Q}(\zeta)$ and L^+ be its maximal real subfield. Since 2 is invertible modulo K, choose m with $2m \equiv 3 - T \pmod{K}$ and put $\beta = \zeta^m Q(\zeta)$. By Lemma 6.1,

$$\overline{\beta} = \zeta^{-m} Q(\zeta^{-1}) = \zeta^{-m+3-T} Q(\zeta) = \beta.$$

Thus β is an algebraic integer in L^+. Multiplication by ζ^m does not change the norm, so

$$\mathrm{N}_{L/\mathbf{Q}}(Q(\zeta)) = \mathrm{N}_{L^+/\mathbf{Q}}(\beta)^2 = M^2.$$

Also $\mathrm{N}_{L/\mathbf{Q}}(\zeta - 1) = \Phi_K(1) = K$. Taking norms of $P(\zeta) = (\zeta - 1)^3 Q(\zeta)$ proves (14). The non-forced case has $P(\zeta) \neq 0$ by Theorem 5.2. □

Corollary 6.3 (Improved no-collision cutoff)**.** *Suppose $K \mid p - 1$. If a non-forced finite-field collision exists, then*

$$K^3 p^2 \leq 12^{K-1}. \tag{15}$$

Consequently $K^3 p^2 > 12^{K-1}$ implies the exact baseline (9).

Proof. A finite-field collision makes P and Φ_K share a root modulo p, so $p \mid \mathrm{Res}(\Phi_K, P)$. Since $K \mid p - 1$, $p \neq K$. Equation (14) then gives $p \mid M$, hence $K^3 p^2 \leq |\mathrm{Res}(\Phi_K, P)|$. On the other hand $|P(\zeta^a)| \leq \|P\|_1 \leq 12$ for all a, giving the upper bound. □

The cutoff is stronger than the elementary $p > 12^{K-1}$ bound, but it remains exponential in K. The next section shows that this is not merely an artifact of a crude norm estimate.

7 An explicit Fibonacci collision slice

Consider

$$c(h^2) - c(h) = c(h) - c(h^{-1}). \tag{16}$$

Since $c(h^{-1}) = -c(h)$,

$$c(h^2) - 3c(h) = -\frac{2(h^2 + 3h + 1)}{(h - 1)(h + 1)}.$$

For odd-order $h \neq 1$ we also have $h \neq -1$, so (16) is equivalent to

$$q(h) = h^2 + 3h + 1 = 0. \tag{17}$$

Theorem 7.1 (Fibonacci resultant identity)**.** *For every odd prime K,*

$$\mathrm{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2. \tag{18}$$

For the corresponding cleared Cayley collision polynomial, the full resultant is $K^3 F_K^2$.

Proof. Let α, β be the roots of $X^2 + 3X + 1$, so $\alpha\beta = 1$, $\alpha + \beta = -3$, and $\Phi_K(X) = (X^K - 1)/(X - 1)$. Therefore

$$\mathrm{Res}(\Phi_K, q) = \frac{(\alpha^K - 1)(\beta^K - 1)}{(\alpha - 1)(\beta - 1)} = \frac{2 - (\alpha^K + \beta^K)}{5}.$$

With $\varphi = (1 + \sqrt{5})/2$, the roots are $-\varphi^2$ and $-\varphi^{-2}$. Since K is odd, $\alpha^K + \beta^K = -L_{2K}$, and the standard identity $L_{2K} = 5F_K^2 - 2$ for odd K gives (18). Comparing with Theorem 6.2, or directly taking the cleared denominator norm, gives $K^3 F_K^2$. □

Because $F_K \sim \varphi^K/\sqrt{5}$, Theorem 7.1 proves that no bound $|M| \leq K^C$ can hold for all collision resultants. Thus any useful polynomial-scale theorem must exploit the arithmetic of eligible prime divisors or directly bound total energy rather than total resultant size.

If p is prime and $K \mid p - 1$, then (16) occurs over $\mathbf{F}_p$ if and only if $p \mid F_K$. Indeed Φ_K splits completely over $\mathbf{F}_p$, and (18) is exactly the common-root criterion.

8 Fibonacci/Lucas ratios form an almost-Sidon set

Assume $q(h) = 0$ and put $t = c(h)$. Substitution $h = (t+1)/(t-1)$ into $q(h) = 0$ gives

$$5t^2 = 1. \tag{19}$$

A direct Binet calculation, equivalently an induction using the standard Fibonacci/Lucas recurrences, gives

$$F_n c(h^n) = L_n c(h), \qquad n \geq 1. \tag{20}$$

Thus the characteristic-zero model for half of the centrally symmetric Cayley set is

$$x_n = \frac{L_n}{F_n}, \qquad S_N = \{\pm x_1, \ldots, \pm x_N\}.$$

Two identities will be used repeatedly:

$$x_m + x_n = \frac{2F_{m+n}}{F_m F_n}, \qquad |x_m - x_n| = \frac{2F_{|m-n|}}{F_m F_n}. \tag{21}$$

Carmichael's primitive-divisor theorem implies that F_j has a prime divisor not dividing any earlier Fibonacci number for every $j > 12$ [10, 11]. We use this only as a divisibility separation principle; the finite range through 12 is checked exactly.

Theorem 8.1 (Exact almost-Sidon energy)**.** *For every $N \geq 3$, if*

$$E^*(S_N) = \sum_{d \neq 0} r_{S_N - S_N}(d)^2,$$

then

$$E^*(S_N) = 2N(4N-3) + 32. \tag{22}$$

The excess over the centrally symmetric forced baseline is exactly 32, independent of N.

Proof. First classify relations between positive x_n.

For equal absolute differences, suppose $m > n$, $r > s$ and

$$\frac{F_{m-n}}{F_m F_n} = \frac{F_{r-s}}{F_r F_s}.$$

If, say, $m > r$ and $m > 12$, a primitive divisor of F_m divides the right cross-multiplied side but none of F_{m-n}, F_r, F_s, a contradiction. Thus beyond the bounded range the upper indices agree. Then F_{m-n}/F_n is strictly decreasing in n, so the lower indices agree.

For equal sums, if $m+n > r+s$ and $m+n > 12$, a primitive divisor of F_{m+n} cannot divide any of F_{r+s}, F_m, F_n, again a contradiction. Thus the index sums agree. With $m+n = r+s = T$, equality reduces to $F_m F_n = F_r F_s$. The identity

$$5F_m F_n = L_T - (-1)^n L_{m-n}$$

then forces the unordered pairs $\{m, n\}$ and $\{r, s\}$ to agree.

Finally $1 \leq x_n \leq 3$. A positive sum can equal an absolute difference only at the endpoint value 2, giving $x_1 + x_1 = x_2 - x_1$. The remaining bounded relations are exactly those involving $x_1 = 1, x_2 = 3, x_3 = 2$:

$$|x_3 - x_1| = |x_2 - x_3| = 1, \quad x_1 + x_1 = x_2 - x_1 = 2, \quad x_1 + x_2 = x_3 + x_3 = 4.$$

The finite residual statement is independently kernel-checked in Lean for indices through 12.

For a centrally symmetric set of size $2N$, forced inversion pairing alone contributes $2N(4N-3)$. The two oriented bins ± 1 have multiplicity 4 rather than two independent multiplicity-2 bins, contributing excess 8 each. The four bins $\pm 2, \pm 4$ have multiplicity 3 rather than a multiplicity-1 and a multiplicity-2 bin, contributing excess 4 each. The total excess is $2 \cdot 8 + 4 \cdot 4 = 32$. □

Remark 8.2. *The broad theorem of Bilu–Hanrot–Voutier [12] supplies primitive divisors for much more general Lucas/Lehmer sequences for sufficiently large index. For Theorem 8.1, the classical real-discriminant Fibonacci theorem of Carmichael already suffices.*

9 A 28-digit exact finite-field certificate

Set

$$K = 131, \qquad p = F_{131} = 1066340417491710595814572169.$$

The repository proves the primality of p by the Lucas criterion using the complete factorization

$$p - 1 = 2^3 \cdot 11 \cdot 89 \cdot 131 \cdot 199 \cdot 521 \cdot 2081 \cdot 9901 \cdot 19801 \cdot 24571$$

and base 3: it checks

$$3^{p-1} \equiv 1 \pmod{p}, \qquad \gcd(3^{(p-1)/q} - 1, p) = 1$$

for every prime q in this factorization.

The explicit integer

$$h = 188706367382658195937453923$$

satisfies

$$h^2 + 3h + 1 \equiv 0 \pmod{p}, \qquad h^{131} \equiv 1 \pmod{p}, \qquad h \neq 1.$$

Since 131 is prime, h has exact order 131. Exact dictionary-based arbitrary-precision computation gives

$$\sum_t C_H(t)^2 = 33442, \qquad (K-1)(2K-5) = 33410, \qquad \text{excess} = 32. \tag{23}$$

The maximum nonzero difference multiplicity is 4. The complete multiplicity histogram is

$$1^{\times 126}, \qquad 2^{\times 8312}, \qquad 3^{\times 4}, \qquad 4^{\times 2}.$$

Independent Python arbitrary-precision and C++20 Boost.Multiprecision implementations verify these numbers and the primality certificate. The finite low-index relation classification in Theorem 8.1 is separately checked by Lean 4.

Status: Finite certificate.. This example is deliberately not extrapolated to an infinite sequence of Fibonacci primes. Its role is to demonstrate that very large exceptional split primes can coexist with only constant excess energy.

For comparison, the older independent design $(p, K) = (461, 23)$ has

$$\sum_t C_H(t)^2 = 1190, \qquad \text{baseline} = 902, \qquad \text{excess} = 288.$$

Thus exceptional reductions can have different collision geometry.

10 The mixed-term target after the Fibonacci obstruction

The preceding results change the natural closure condition. A theorem saying "all non-forced collisions disappear once p is polynomially large in K" is neither the only nor the most relevant target. Equation (4) shows that what ultimately matters is the L^2 contribution of C_H, hence the energy (8).

The next quantitative problem is therefore to prove a uniform or sufficiently averaged bound for

$$\sum_t C_H(t)^2$$

for $A = c(H \setminus \{1\})$ in a polynomial-scale range of K. Candidate routes include incidence/sum-product estimates for Mobius images of multiplicative subgroups, arithmetic classification of high-multiplicity differences, and structured-plus-pseudorandom decompositions. Before importing any generic energy theorem, the required exponent must be propagated through (4) and (6); a numerically attractive bound that is too weak after the $(p-2)$ factor is not sufficient.

Status: Open.. The new square-resultant and Fibonacci theorems remove one incorrect proof target and replace it with a sharper quantitative one; they do not themselves close the mixed contribution in the desired moment regime.

11 The off/off elliptic route

After separating the mixed term, the unresolved contribution is the centered energy of $O(t)$. Historical symbolic work organizes the relevant fibers as a one-parameter genus-one family and suggests a quartic/Weierstrass model, a nonconstant rational j-map, visible bad fibers and same-j parameter symmetries.

Status: Theorem draft.. A publication-grade proof must re-establish or certify all of the following for the actual family:

1. a smooth projective model on an explicit open base;
2. the complete discriminant/bad-fiber locus, including characteristic exceptions;
3. affine/projective boundary corrections;
4. purity, lissity, geometric irreducibility, and uniform conductor/ramification control of the relevant R^1 sheaf;
5. all geometric isomorphisms and twists between parameter fibers;
6. separable and inseparable isogeny exclusions or explicit exceptional strata;
7. the exact hypotheses of the trace-sheaf correlation theorem used.

Michel-type family arguments [13], quantitative sheaf theory [14], Tate/isogeny technology [15], and the modern joint-family example of Grove–Saad [16] are relevant tools, but none is a plug-in proof for this pencil.

Conditional on the required square-root correlation with uniform constants, the existing combinatorics suggest

$$O\Big(K^2 p + K^4 p^{1/2} + K^4\Big), \tag{24}$$

with intended regime $K = o(p^{1/4})$. Equation (24) remains a draft conclusion.

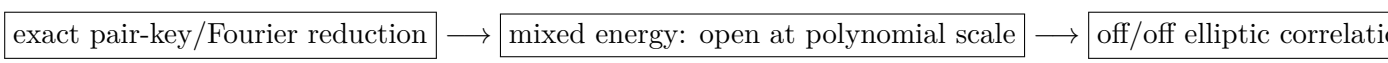
12 Finite spectral diagnostics and the microscopic BGS gap

The maintained finite spectral code diagonalizes the full complex Hermitian sector matrix

$$\mathcal{H}(q) = M(q) + M(q)^*.$$

A reconstruction-era implementation briefly replaced it by its entrywise real part; that bug was removed and guarded by regression. Current spacing records are descriptive empirical-CDF comparisons only. Pooled unfolded gaps are dependent, so no IID Kolmogorov–Smirnov significance is attached. The forward-window statistic retained in the repository is likewise not stationary number variance.

The logical chain is



Closing an earlier box does not close a later one. A local-statistics theorem still requires an explicit comparison/local-law mechanism, or a genuinely arithmetic replacement, at the mean-spacing scale with the correct symmetry class and uniformity.

13 Reproducibility and independent failure modes

The maintained verification stack is deliberately heterogeneous:

- Python/SymPy: transparent symbolic algebra, exact resultants, finite-field reference computations, primality and energy certificates;
- C++20: independent exact/exhaustive kernels and the 28-digit Boost.Multiprecision Fibonacci witness;
- Lean 4: kernel-checked pair-key certificates, fixed Cayley controls, torus facts, historical falsifiers, and the bounded Fibonacci-ratio residual;
- TeX/CI: publication-facing integration and independent manuscript build.

The C++ arbitrary-precision verifier intentionally uses a different code path from Python. The Lean module does not formalize Carmichael's theorem; it certifies exactly the finite range left after the external primitive-divisor theorem is applied. This separation keeps the evidence graph honest.

The current tree contains no Julia sources. Historical Julia files remain accounted for in repository provenance; false algorithms were not ported merely to increase language diversity.

14 Conclusion

The exact pair-key/Fourier reduction and Cayley transform isolate the fourth-moment arithmetic obstruction sharply. Prime-order cotangent rigidity removes characteristic-zero accidental collisions. The new norm identity

$$\mathrm{Res}(\Phi_K, P) = K^3 M^2$$

explains the systematic square structure seen in fixed-K computations and strengthens the worst-case cutoff. The Fibonacci slice then gives a more important lesson: large exceptional norms and even very large exceptional split primes do not force large additive energy. Its exact rational energy is only baseline plus 32, and the 28-digit $K = 131$ certificate reproduces that structure in finite characteristic.

This changes, but does not close, the BGS programme. The next mixed-term problem is a quantitative energy theorem for Mobius images of multiplicative subgroups. The off/off elliptic correlation remains a separate geometric problem. Even success on both fronts would still leave the microscopic BGS bridge as an independent local-statistics problem.

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